

POLICY BRIEF

What Should a Third East African Country Learn from Kenya and Rwanda's Divergent Digital Public Infrastructure Paths?

Replicable Lessons, Context-Specific Constraints, and a Sequenced Hybrid Strategy for Tanzania

Lisa Uwonkunda | Independent Analyst | Nairobi, Kenya | June 2026

Note on this version: the composite scores below have been corrected from an earlier draft in which Kenya's and Rwanda's composite calculations omitted sub-indicator 3.1 (National Digital ID Coverage), while Tanzania's correctly included it. All figures here reflect the full 12-indicator calculation, verified directly against the underlying SQL scoring data.

Executive Summary

Rwanda leads on composite DPI capability (6.82/10) despite a GNI per capita 45% lower than Kenya's (6.11/10), confirming that execution fidelity, the consistent conversion of digital mandates into functional public goods, matters more than income level. Tanzania scores 3.51/10, but the composite alignment gap of 2.95 (nearly four times Kenya's 0.76) reveals the deeper problem: Tanzania's strategy is more ambitious relative to its current capacity than either reference model. Tanzania does not need a new model. It needs a sequenced strategy that builds enabling conditions before mandating outcomes.

This brief examines Kenya and Rwanda as two structurally distinct DPI development models and derives actionable lessons for Tanzania through an original Digital Public Infrastructure Readiness Index, a 12 sub-indicator, four-dimension scoring framework applied consistently across all three countries with a December 2023 primary anchor date. The central argument is that neither model transfers wholesale to Tanzania. Instead, specific elements can be copied, others must be adapted, and some cannot be replicated without enabling conditions Tanzania has not yet established.

Key Findings

- Rwanda's zero alignment gap on Digital Public Stack Maturity (D3: 9/10 capability, 9/10 ambition) is the quantitative signature of execution fidelity, the only dimension where any country's ambitions are fully matched by demonstrated capability.
- Tanzania's three largest alignment gaps, which are Government Services Digitized (gap: 5), Regulatory Maturity (gap: 4), and Political Commitment Durability (gap: 4), all have LOW or CONDITIONAL transferability, meaning Tanzania cannot simply copy either model to close them.

- Tanzania's E-Participation Index collapsed from 0.618 (Rank 67) in 2018 to 0.288 (Rank 137) in 2024, a regression that quantifies the sequencing risk that infrastructure investment without trust-building produces declining citizen engagement.
- Tanzania already has 63.2 million active mobile wallets and an operational TIPS interoperability switch, placing it further along the market-led payment infrastructure path than the standard narrative assumes.

Recommendation

A three-move sequenced hybrid strategy:

- Close the regulatory enforcement gap and operationalise TIPS before any adoption mandate.
- Centralise service delivery using Rwanda's binding agency compliance mechanism, not Kenya's voluntary approach.
- Embed DPI in revenue-generating, open-source structures that survive political cycles Tanzania cannot guarantee through continuity alone.

1. The Composite DPI Readiness Index

Country	Composite Capability	Composite Ambition	Alignment Gap
Rwanda	6.82	8.47	1.65
Kenya	6.11	6.87	0.76
Tanzania	3.51	6.46	2.95

Composite scores are weighted averages across all 12 sub-indicators per the project's scoring rubric. Interactive visualizations, including the full dimension and sub-indicator breakdown, are available via the project's Tableau dashboard (linked in Data Sources, Section 6).

Rwanda leads on composite capability despite its smaller economy because its DPI model concentrates execution on specific high-priority domains rather than pursuing broad e-government development. Kenya's narrowest alignment gap (0.76) reflects realistic target-setting rather than exceptional performance. Tanzania's gap of 2.95 is not simply a resource problem. It is a sequencing problem. Its Digital Economy Strategic Framework 2024–2034 sets targets that require enabling conditions Tanzania has not yet built.

2. Two Structurally Distinct DPI Models

Kenya: Market-Led, Ecosystem-Driven

Kenya built its DPI on a private sector asset that predated any government digital strategy. M-Pesa's 2007 launch created payment rails that 80% of the population was already using before the state had a coherent digital agenda. The state's initial role was permissive non-regulation, a deliberate policy choice that allowed private sector experimentation before formalisation. The 2013 Presidential directive and the ICT Authority's enforcement of the Government Enterprise Architecture came after M-Pesa had already established the ecosystem foundation, making the state a standardiser of an existing stack rather than its architect. Kenya's model is replicable where a dominant private sector player exists and can be incentivised to take infrastructure risk. It fails where the private sector is fragmented.

Rwanda: State-Led, Mandate-Driven

Rwanda's model rests on what this analysis terms execution fidelity: the government's demonstrated, multi-year track record of converting digital mandates into functional public goods. IremboGov was built and mandated before significant private sector demand existed; agencies had to comply, producing a portal that processes 500,000 applications monthly across 100+ service categories. Rwanda's compact population (14.7 million) made enforcement tractable. Its post-1994 political settlement provided continuity across Vision 2020 and Vision 2050 that no policy design can replicate. Rwanda's model is replicable where state authority is broad, population is manageable, political commitment is durable, and external funding is available. It fails where any of these conditions is absent.

The Structural Contrast

Kenya built DPI on top of private sector infrastructure that preceded state strategy. Rwanda built private sector participation into a state-designed architecture after the fact. A country choosing between these models is not choosing between better and worse outcomes. It is choosing between different enabling conditions, and the choice is only available if those conditions exist.

3. Evidence by Dimension

Dimension	Kenya Capability	Rwanda Capability	Tanzania Capability	Key Finding
D1: Infrastructure Foundation (wt: 25%)	6.44	5.08	4.44	Tanzania's infrastructure score (70.8 GSMA) nearly matches Kenya. Affordability is the binding constraint
D2: State Capacity for Digital Gov. (wt: 30%)	5.67	6.33	3.00	Tanzania PDPA enforcement commission not yet operational — 2/10 regulatory maturity
D3: Digital Public Stack Maturity (wt: 30%)	6.50	9.00	3.50	Rwanda's zero alignment gap on D3 is the quantitative signature of execution fidelity
D4: Political Economy & Sustainability (wt: 15%)	6.00	6.33	3.00	Tanzania strategy too new to assess durability — 3/10. Donor dependency is a shared constraint across all three countries

Finding 1: Rwanda's D3 score of 9/10 with zero alignment gap

Rwanda is the only country in this analysis where capability and ambition are perfectly matched on any dimension. This is not because Rwanda's ambitions are modest; NST2 targets are demanding. It is because Rwanda's institutional machinery converts targets into outcomes at a rate no comparator achieves. For Tanzania, this is not replicable by strategy design; it is the output of decades of consistent delivery.

Finding 2: Kenya's D3 patchy performance (6.5/10) despite 22,000+ eCitizen services

Kenya lists over 22,000 services on eCitizen. Rwanda lists 100+. Rwanda's 9/10 versus Kenya's 6.5/10 on the same dimension reveals the difference between listing and delivering. Kenya's voluntary agency compliance model means service quality is

inconsistent. This directly informs Tanzania's approach: copy Rwanda's binding compliance mechanism, not Kenya's service count.

Finding 3: Tanzania's E-Participation collapse

Tanzania's E-Participation Index fell from 0.618 (Rank 67) in 2018 to 0.288 (Rank 137) in 2024, the sharpest regression in the sub-region. Tanzania has been building digital systems without building digital trust. Infrastructure investment without citizen engagement produces declining adoption. This is the quantitative case for closing the enforcement gap before mandating adoption.

Finding 4: Rwanda's D1 alignment gap of 3

Rwanda scores 5.08 on infrastructure capability but 8.0 on ambition, a gap of 3, the largest infrastructure gap of any country. Rwanda's DPI model succeeded despite infrastructure constraints by substituting state mandate for market-driven adoption. Tanzania, with a higher infrastructure capability (4.44) but weaker mandate capacity, cannot replicate this substitution.

4. Tanzania: Sub-Indicator Detail and the Alignment Gap

Code	Sub-Indicator	Capability	Ambition	Gap
1.1	Mobile Internet Penetration	4	6	2
1.2	Mobile Money Penetration	6	8	2
1.3	Smartphone Penetration	3	5	2
2.1	E-Government Development Index	4	6	2
2.2	Regulatory Maturity	2	6	4
2.3	Government ICT Budget Percent	3	5	2
3.1	National Digital ID Coverage	3.5	7.5	4
3.2	Government Services Digitized	2	7	5
3.3	Payment Interoperability	5	8	3
4.1	Domestic vs Donor Funding Ratio	3	5	2
4.2	Private Sector Co-Investment	3	6	3
4.3	Political Commitment Durability	3	7	4

Every sub-indicator where the ambition score exceeds the capability score represents an area where Tanzania's stated ambitions exceed its current capability, an implementation risk. See the project's Tableau dashboard for the full interactive capability-versus-ambition visualization, color-coded by transferability classification (Section 5).

5. What Tanzania Can Copy, Adapt, and Must Build Independently

Transferability	Sub-Indicators	Tanzania Action
HIGH — Copy directly	Mobile internet (1.1), Smartphone penetration (1.3), Private sector co-investment framework (4.2)	Competitive telco licensing, remove device import duties, establish digital PPP framework using Vodacom/Airtel as entry points
CONDITIONAL —	Mobile money (1.2), EGDI (2.1),	Copy Rwanda's compliance

Adapt with sequencing	Digital ID (3.1), Services digitized (3.2), Payment interoperability (3.3)	mechanisms not Kenya's voluntary approach, sequence NIDA physical delivery before digital mandate, TIPS before G2P digitization
LOW — Cannot be transplanted	ICT budget % (2.3), Domestic funding ratio (4.1), Political durability (4.3)	Mitigate through revenue-generating DPI design, open-source architectures, escalating co-financing requirements in multilateral agreements

6. A Sequenced Hybrid Strategy for Tanzania

Tanzania's DPI path must be sequenced to build enabling conditions deliberately such as trust infrastructure before adoption mandates, payment interoperability before service digitization, open architectures before scale. Rwanda's mandate model onto a population four times larger with institutional foundations that have not yet demonstrated Rwanda's level of delivery fidelity.

Move 1: Build Trust and Payment Foundation (0–18 months)

Tanzania's PDPA enforcement commission must demonstrate credibility through documented enforcement, not just establishment. At least three enforcement actions against significant data controllers before any digital service mandate is issued. In parallel, TIPS must move from technical availability to market habit through active incentivisation of merchant and government adoption. Tanzania already has 63.2 million active mobile wallets. The regulatory coordination that makes it an interoperable public infrastructure does not yet fully function. This move borrows from Kenya's legal scaffolding approach but demands the enforcement follow-through Kenya itself has struggled to achieve.

Move 2: Centralise Services with Binding Compliance (18 months–4 years)

Once trust infrastructure is credible and TIPS is achieving network effects, Tanzania's highest-leverage move is consolidating its 898 fragmented service systems into a unified portal, using Rwanda's binding agency compliance mechanism. Kenya's eCitizen experience is the cautionary case: 22,000+ services listed, inconsistent quality, because agencies had no penalty for non-migration. Rwanda's IremboGov works because compliance was non-negotiable. Tanzania's portal must be built with a legal mandate for agency migration, a specific timeline, penalties for non-compliance, and a citizen-centric design requirement like Kiswahili-first, mobile-first, functional on 2G/3G, that neither Kenya nor Rwanda fully achieved because neither was designed for Tanzania's specific conditions.

Move 3: Embed DPI in Durable Structures (Ongoing from Year 1)

Tanzania's Digital Economy Strategic Framework (2024–2034) is too new to have survived a government transition. Three structural mechanisms should begin from Year 1 rather than waiting for the strategy to prove its durability:

- Make DPI revenue-generating through G2P digitization savings and service fee structures so any future government has a fiscal incentive to maintain it, not merely a political one.
- Build in open-source, in-house architectures (Tanzania has already started with TIPS) to avoid vendor dependency that makes reversal cheap and easy.
- Anchor key investments in multilateral lending arrangements with domestic co-financing requirements that escalate over time, making budget commitment contractual rather than discretionary.

7. Methodology and Data Notes

Index Design

This analysis employs an original Digital Public Infrastructure Readiness Index comprising 12 sub-indicators across four dimensions, weighted to reflect the relative importance of each layer in DPI development theory. All scoring rubrics were defined before data collection to prevent retrospective calibration. Capability scores (1–10) reflect current demonstrated performance; ambition scores (1–10) reflect stated targets in national strategy documents. The alignment gap (ambition minus capability) identifies implementation risk areas, the primary analytical contribution of this framework relative to existing indices.

Anchor Date and Data Vintage

The primary anchor date is December 2023. All figures cited reflect the most recent data available as of that date. The GSMA Mobile Connectivity Index data cited (2025 vintage) is noted explicitly. Estimated figures are marked throughout and estimation logic is documented in the source database.

Data Limitations

- Tanzania's ICT budget is not separately reported as a clean line item as it is distributed across eGA, TCRA, and ministry allocations. Fragmentation is itself treated as evidence of low prioritization.
- Mobile money penetration figures vary by measurement definition across sources (active-in-last-90-days versus ever-registered); comparisons use the most directly comparable, regulator- or survey-published figures available, with the specific definition noted in the source database.
- Qualitative sub-indicators (2.2, 3.3, 4.2, 4.3) reflect the author's analytical judgment applied consistently against pre-defined rubric thresholds.

Data Sources

Source	Sub-Indicators
UN e-Government Survey 2024	2.1 E-Government Development Index (Kenya: 0.6314, Rwanda: 0.5799, Tanzania: 0.4327)
GSMA Mobile Connectivity Index 2025	1.1 Mobile Internet Penetration (Kenya: 59.9, Rwanda: 54.4, Tanzania: 49.2)
Communications Authority Kenya / FinAccess 2024	1.2 Mobile Money Penetration Kenya (75.1% active)
GSMA Intelligence / DataReportal 2024	1.3 Smartphone Penetration (estimated from MCI data)
World Bank ID4D Database	3.1 Digital ID Coverage (Rwanda 92%, Kenya 88% est., Tanzania 23% physical)
IremboGov / eCitizen / eGA portals (manual count)	3.2 Government Services (Rwanda 100+, Kenya 22,000+, Tanzania 0 citizen-facing)
Rwanda NIDA / Kenya ODPC / Tanzania TCRA	2.2 Regulatory Maturity (enforcement action)

	counts)
Rwanda MINECOFIN / Kenya Treasury / Tanzania MoF	2.3 Government ICT Budget (estimated)
Rwanda BNR / Kenya CBK / Tanzania BOT	3.3 Payment Interoperability
World Bank ODA data / national budget documents	4.1 Domestic vs Donor Funding Ratio (estimated)
Rwanda RDB / Kenya PPP Authority / Tanzania PPRA	4.2 Private Sector Co-Investment
National strategy documents (Rwanda Vision 2050, Kenya Digital Master Plan 2022-2032, Tanzania DESF 2024-2034)	4.3 Political Commitment Durability

Interactive dashboard and full SQL database schema and source documentation available in this repository.

This policy brief was produced as an independent analytical project. All scoring judgments and recommendations represent the author's analytical conclusions and do not reflect the position of any institution. Data limitations are documented above. The author welcomes challenge and correction.